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A Rapid Information-Theoretic Hybrid Algorithm for Mixed-Integer Optimization

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Abstract We present a method for approximating solutions to large-scale mixed-integer programs by combining: (i) entropy-regularized linear relaxation solved via Sinkhorn iterations, and (ii) structured integerization via minimum Kullback–Leibler divergence rounding. The approach leverages information-theoretic projections for scalability while producing high-quality solutions through entropy-aware rounding. The method applies to a host of problems, including transport, network flows, facility location, set covering, and population synthesis. For lattice polytopes (e.g., transportation and assignment), it recovers exact optimal solutions. For population synthesis, it achieves zero integrality gap on totally unimodular systems and provably bounded deviations for more complex non-TU linked-constraint cases. For general set covering, it provides high-quality approximate solutions with the classical logarithmic approximation guarantee. We evaluate on three representative problem classes at varying scales: (i) population synthesis achieving exact marginals in under 2 s for 160×160 tables, (ii) transportation problems solved optimally in under 1 s for 300×300 instances, and (iii) synthetic set covering across three scales (100×3000 to 300×5000), where the hybrid reaches high-quality feasible solutions within about 2 s and its advantage over time-limited MIP becomes more pronounced as instance size grows, with the larger scales retaining a positive gap advantage throughout a 30 s horizon. The method bridges information geometry and combinatorial optimization, providing a scalable, theoretically grounded alternative to traditional MIP heuristics.

Keywords Entropy regularization · Mixed-integer programming · KL projection · MAP rounding · Hybrid optimization

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1 Introduction

Large-scale mixed-integer programs (MIPs) present a fundamental computational trade-off: exact methods guarantee optimality but scale poorly on large instances, while heuristics scale well but sacrifice solution quality and theoretical guarantees. This paper bridges two historically distinct research streams—matheuristics that exploit continuous relaxations [14, 13, 10, 5] and information-theoretic projection methods [12, 8, 9]—by showing how entropy regularization can simultaneously accelerate relaxation solving and guide principled integerization.

We introduce an algorithm which, for simplicity, we refer to as RICH (Rapid Information-TheoretiC Hybrid; the digraph "IC" provides the "C").[1] It provides an alternative to standard linear programming (LP) relaxation by using an entropy-regularized convex program solved via Iterative Proportional Fitting (IPF) or Sinkhorn scaling [11, 34]. Integer solutions are then recovered through minimum Kullback–Leibler (KL) divergence rounding. Unlike branch-and-cut or primal heuristics that alternate between discrete and continuous subproblems, RICH executes a single information-theoretic projection: near-linear time relaxation solving followed by structured rounding whose cost is low-degree polynomial in the problem size.

This fusion yields several complementary properties. On totally unimodular (TU) structures—transportation, assignment, and unimodular population synthesis—the entropy-regularized relaxation recovers exact optimal integer solutions. For non-TU cases, the method provides problem-specific guarantees: bounded deviations for linked-constraint synthesis and a logarithmic approximation guarantee for set covering, together with strong empirical performance. In both cases, the entropy-aware rounding produces statistically interpretable solutions while maintaining computational efficiency.

1.1 Related work: heuristics and matheuristics for MIP

Primal heuristics have long been central to the practical success of MIP solvers, providing early feasible incumbents and enabling anytime use [5]. Classical categories include rounding and diving heuristics (iteratively solving LPs and fixing variables), improvement heuristics (locally refining feasible solutions), and relaxation-based neighborhood searches, all embedded in solvers such as SCIP [1] and Gurobi [19].

Two historically influential schemes are *Local Branching* and *RINS*. Local Branching [14] adds a cardinality constraint around a feasible solution to explore a small subspace with strong improvement potential. RINS (Relaxation Induced Neighborhood Search) [10] fixes variables where the LP relaxation and incumbent agree, then re-optimizes the reduced problem. Both have inspired a broad family of matheuristics coupling exact search with heuristic guidance.

The *Feasibility Pump* (FP) [13, 16] alternates between solving a continuous relaxation and projecting to integrality to find feasible solutions rapidly. FP emphasizes speed over solution quality; RICH takes a complementary stance—prioritizing

[1] The acronym is approximate: "Rapid Information-TheoretiC Hybrid" yields RITCH, and we use RICH for brevity. We retain the name for continuity across the entropy-MIP paper series.

solution quality via entropy regularization and dual guidance while retaining polynomial-time scalability.

A property that distinguishes RICH from most primal heuristics is the *guaranteed feasibility* of its output: the pipeline always terminates with an integer feasible solution, regardless of instance hardness or time budget. Feasibility Pump, RINS, and Local Branching can fail to find any feasible solution within a fixed budget on hard instances; RICH cannot. This guarantee is structural rather than empirical—it follows from the feasibility-preserving design of the drop–fix stage and the greedy repair embedded in the rounding step (Section 2.7).

Other proximity-based heuristics tighten or re-optimize around LP solutions, including Proximity Search [15], RENS (Relaxation Enforced Neighborhood Search) [5], and structure-driven fix-and-propagate [18]. Large Neighborhood Search (LNS) and its adaptive variant ALNS [28] follow a similar destroy-and-repair philosophy but rely on exact or constraint-programming (CP) repair engines. These techniques collectively define the matheuristics landscape surveyed in [23].

A parallel development is the rise of *learning-augmented heuristics*, where machine learning guides branching, cutting, or diving decisions [21, 4]. These methods reduce time to strong incumbents and generalize across instance families, complementing classical search-based heuristics.

1.2 Contributions and positioning

RICH differs from existing heuristics and matheuristics in how it constructs and exploits relaxations:

1. It replaces LP relaxations with an *entropy-regularized* convex formulation solved via fast KL I-projection algorithms (IPF/Sinkhorn), achieving near-linear iteration cost per projection step.
2. It performs *structured integerization* by minimizing KL divergence from the fractional solution to the integer feasible set, yielding theoretically grounded rounding with provable convergence properties.
3. On TU-structured classes (assignment, transport), it provides a polynomial-time exact pipeline. On more general classes, including set covering and non-TU population-synthesis variants, it provides theoretically grounded approximation guarantees together with competitive empirical time-quality trade-offs on the tested instances. This trade-off is especially relevant when the method is embedded in repeated decision pipelines, where short and stable runtimes may matter as much as closing the final few percent of optimality gap.

In spirit, RICH is a matheuristic operating in the geometry of information rather than Euclidean distance. It is compatible with learning-augmented strategies and dual guidance, and extends the toolbox of primal heuristics toward scalable, information-theoretically grounded optimization. A natural further direction, beyond the scope of this paper, is to exploit the Boltzmann-like probability distribution induced by the entropy relaxation to derive principled uncertainty estimates for near-optimal solutions—a connection we leave for future work.

2 Methodology

At first glance, what we present may appear counterintuitive: we deliberately turn a linear relaxation into a *nonlinear* optimization problem by adding an entropy penalty. This suggests increasing complexity from a solver perspective. Here, however, the entropy term induces a powerful structure because the resulting relaxation reduces to a *KL projection problem*, solvable by highly efficient IPF or Sinkhorn-type scaling methods. These are multiplicative updates, closed-form, that scale nearly linearly with problem size and converge rapidly. In short, the nonlinear problem is, paradoxically, easier to solve at scale than the LP relaxation for problems with the appropriate structure. A second benefit is that the entropy relaxation recasts the problem in probabilistic terms, providing natural guidance for the subsequent rounding stage. Rather than applying ad hoc heuristics to fractional values, we interpret the solution as a probability distribution and select the most likely integer realization.

2.1 Convex relaxation via entropy regularization

The central idea of our hybrid approach is to enrich the standard MIP relaxation with an entropy term. We start from the integer program

$$\text{(MIP)} \quad \min_{x \in \mathbb{Z}_+^n} \; c^\top x \qquad \text{s.t. } Ax = b,\ \ell \le x \le u, \tag{2.1}$$

whose linear relaxation is

$$\text{(LP)} \quad \min_{x \ge 0} \; c^\top x \qquad \text{s.t. } Ax = b,\ \ell \le x \le u. \tag{2.2}$$

We introduce the entropy-regularized formulation

$$\text{(Hybrid)} \quad \min_{x \ge 0} \; c^\top x + \tau \sum_{i=1}^{n} x_i(\log x_i - 1) \qquad \text{s.t. } Ax = b,\ \ell \le x \le u, \tag{2.3}$$

with smoothing parameter $\tau > 0$. Small τ recovers the sharp LP relaxation (2.2), while larger values promote interior, well-conditioned solutions. Here $x = (x_1, \ldots, x_n)^\top$ denotes the decision vector. We denote by x_τ the optimal solution to (2.3) for a given τ.

2.1.1 Why entropy rather than LP?

The entropy relaxation serves three purposes that distinguish it from the standard LP relaxation.

First, the LP may produce sparse extreme-point solutions with many zero entries, making probabilistic interpretation undefined: one cannot compute $-\log x_i$ when $x_i = 0$. The entropy term $\tau \sum_i x_i(\log x_i - 1)$ acts as a logarithmic barrier that pushes solutions into the strict interior, ensuring $x_\tau > 0$ everywhere and enabling the probabilistic rounding of Section 2.2.

Second, the entropy-regularized problem admits fast iterative solution via Sinkhorn/IPF scaling. Per-iteration cost is $O(\text{nnz}(A))$, matching the cost of a single sparse

matrix-vector product. For dense instances, simplex and interior-point methods require $O(n^3)$ per pivot or barrier iteration; IPF avoids this by construction. On sparse structured problems, both approaches can be competitive, but IPF provides a particularly clean closed-form update rule that requires no factorization.

Third, and most importantly, the entropy formulation provides a *statistical interpretation*: the solution x_τ can be seen as a Boltzmann-weighted probability distribution over the constraint set, parameterized by the cost vector c and temperature τ. This enables principled MAP rounding with provable convergence properties as $\tau \to 0$ (Theorems 2.1 and 2.2).

To summarize the trade-off: relative to the LP relaxation, the entropy formulation gives up a small amount of bound quality (the τ penalty slightly loosens the relaxation) but gains a strict-interior solution suitable for probabilistic rounding, and a rigorous statistical framework that underpins the zero-gap results for integral polytopes. These advantages are not available from the LP relaxation directly.

2.1.2 Projection formulation and solution via IPF/Sinkhorn

The *Bregman divergence* associated with $\phi(x) = \sum_i x_i(\log x_i - 1)$ is $D_\phi(x\|q) = \sum_i [x_i(\log x_i - \log q_i) - x_i + q_i]$, which reduces to the Kullback–Leibler (KL) divergence when $\sum_i x_i = \sum_i q_i$. Using $q_i := e^{-c_i/\tau}$, the objective (2.3) is equivalent (up to constants) to minimizing $D_\phi(x\|q)$ over the feasible set. When $\sum_i x_i$ is fixed by the constraints, this reduces to the KL I-projection:

$$\min_{x \geq 0} \ \mathrm{KL}(x\|q) \quad \text{s.t. } Ax = b,\ \ell \leq x \leq u, \qquad q_i = e^{-c_i/\tau}. \tag{2.4}$$

For the box-unconstrained case ($\ell = 0$, $u = \infty$), the first-order conditions yield the closed-form exponential-family solution $x_i = q_i \exp(-(A^\top \lambda)_i/\tau)$, where λ are the Lagrange multipliers for $Ax = b$. These conditions correspond exactly to multiplicative IPF/Sinkhorn updates [34, 8, 9]. When finite box constraints $\ell \leq x \leq u$ are present, they are handled via coordinate-wise projections within Dykstra's alternating scheme [12]. For problems without fixed mass (e.g., set covering), the full Bregman form is used rather than KL, but the solution method remains identical.

Connection to spatial interaction models. The entropy-regularized relaxation has close parallels to gravity models in transportation research [36, 26], where entropy maximization subject to flow conservation yields solutions of the form $T_{ij} = A_i O_i B_j D_j \exp(-\gamma c_{ij})$. Our approach embeds costs into the prior $q_i = e^{-c_i/\tau}$ rather than imposing them as budget constraints, but the mathematical machinery is identical.

2.2 MAP rounding

The second stage converts the fractional solution x_τ into a feasible integer solution. For *matrix-structured problems* (assignment, transportation), the solution takes the form of a matrix X_τ where $X_\tau(i,j)$ denotes the probability of assigning item i to position j. MAP rounding is described here for this matrix-structured case, which is the canonical setting. For general MIPs without this structure, we use dual-guided or randomized rounding as described in Section 2.7; the probabilistic interpretation carries over but the rounding subproblem changes form.

From probabilities to costs via entropy maximisation. $X_\tau(i,j)$ is the entropy-maximised probability for assignment (i,j) (by the Birkhoff–von Neumann theorem, X_τ is a convex combination of permutation matrices — that is, a meso-state in the sense of [36], with each permutation matrix representing a micro-state). The most likely integer realisation is therefore the permutation that maximises $\prod_i X_\tau(i,\sigma(i))$. By the same argument that justifies maximising $\log W\{T_{ij}\}$ in entropy-maximisation models [36] — a monotone transformation leaves the maximiser unchanged — this is equivalent to minimising $-\sum_i \log X_\tau(i,\sigma(i))$. Since the entropy barrier ensures $X_\tau > 0$ everywhere (Section 2.1), we define $C_{ij} = -\log X_\tau(i,j)$ and solve:

$$P^* \in \arg\min_{P\in\mathcal{P}_n} \sum_{i,j} P_{ij}\,(-\log X_\tau(i,j)). \tag{2.5}$$

This reduces MAP rounding to a standard linear assignment problem, solvable in $O(n^3)$ time via the Hungarian or Auction algorithms.

Example. Suppose the entropy relaxation for a 3×3 assignment yields

$$X_\tau = \begin{pmatrix} 0.8 & 0.1 & 0.1 \\ 0.2 & 0.7 & 0.1 \\ 0.1 & 0.2 & 0.7 \end{pmatrix}.$$

Each row represents a probability distribution over assignments for one item. The negative log-likelihood cost matrix is

$$C = -\log X_\tau \approx \begin{pmatrix} 0.22 & 2.30 & 2.30 \\ 1.61 & 0.36 & 2.30 \\ 2.30 & 1.61 & 0.36 \end{pmatrix}.$$

High probabilities (0.7–0.8) yield low costs (< 0.4), while low probabilities (0.1) incur high costs (> 2).

A feasible integer assignment must assign each item to exactly one position and each position to exactly one item—a permutation σ. With three items there are $3! = 6$ candidates. MAP rounding evaluates the total cost $\sum_i C_{i,\sigma(i)}$ for each:

Assignment σ	Cost	Probability
$(1\to1,\ 2\to2,\ 3\to3)$	$0.22+0.36+0.36=0.94$	$0.8\times0.7\times0.7=0.392$
$(1\to1,\ 2\to3,\ 3\to2)$	$0.22+2.30+1.61=4.13$	$0.8\times0.1\times0.2=0.016$
$(1\to2,\ 2\to1,\ 3\to3)$	$2.30+1.61+0.36=4.27$	$0.1\times0.2\times0.7=0.014$
$\vdots$	$\vdots$	$\vdots$

The diagonal assignment achieves the minimum cost 0.94 and the highest probability 0.392; every other permutation incurs cost above 4.

The intuition of MAP rounding is this. The entropy relaxation produces a soft assignment matrix, where each row spreads probability mass across several candidate matches rather than making a single discrete choice. MAP rounding then converts this soft object into one valid assignment by selecting the permutation matrix that is most consistent with those probabilities. Instead of rounding entries independently, which could violate the one-to-one structure, it solves a standard assignment problem to recover the single globally consistent matching most strongly supported by the relaxed solution.

2.3 Zero-gap theorem for problems with integral LP relaxations

For problems whose LP relaxation automatically yields integer solutions—a class that includes all totally unimodular systems—MAP rounding is not merely heuristic: in the limit $\tau \downarrow 0$ it recovers the true combinatorial optimum.

Theorem 2.1 (Zero-gap property) *Consider the integer program* $\min\{\langle c, x\rangle : Ax = b,\, x \in \mathbb{Z}^n_{\geq 0}\}$ *where the constraint polytope* $\mathcal{P} = \{x \geq 0 : Ax = b\}$ *has only integer vertices. Let* x_τ *solve the entropy-regularized relaxation*

$$\min_{x \geq 0} \langle c, x\rangle + \tau \sum_i x_i(\log x_i - 1) \quad s.t. \quad Ax = b,$$

and let x^*_τ *denote its MAP-rounded integer solution. Then*

$$\lim_{\tau \downarrow 0} \langle c, x^*_\tau\rangle = \min\{\langle c, x\rangle : Ax = b,\, x \in \mathbb{Z}^n_{\geq 0}\}.$$

If the LP optimum is unique, then $x^*_\tau \to x^\star$ *pointwise along a subsequence.* Note: uniqueness of the optimum implies strict complementarity (non-degeneracy), a condition that may fail for highly symmetric instances.

Proof See Appendix B.

The intuition behind Theorem 2.1 is simple. The entropy term smooths the optimization problem and spreads probability mass across feasible solutions when τ is positive. As τ decreases, this smoothing weakens and the mass concentrates increasingly on the cheapest feasible assignment. On problem classes whose LP relaxation is already integral, that concentration occurs around the true integer optimum itself, so the MAP rounding step eventually recovers exactly the optimal discrete solution.

Remark 2.1 Important special cases satisfying the integrality condition include:

- *Totally unimodular (TU) systems:* When A is TU and b is integral, all vertices of $\mathcal{P}$ are integral [33]. This covers assignment, transportation, network flows, and bipartite matching.
- *Lattice polytopes:* More generally, any bounded polyhedron whose vertices all have integer coordinates satisfies the condition.

For these problems, entropy relaxation with MAP rounding provides a polynomial-time exact algorithm.

The same reasoning extends beyond square assignment to rectangular assignment, multi-marginal optimal transport, and any problem whose LP relaxation automatically yields integer solutions. In these cases, entropy relaxation with MAP rounding exactly recovers the optimal solution.

2.4 Convergence rates for lattice polytopes

Theorem 2.1 establishes asymptotic exactness but leaves open practical questions: *How fast does convergence occur, and at what temperature τ can exact recovery be expected?*

Understanding finite-temperature behavior is essential because numerical precision limits how small τ can be before exponential weights $e^{-c_i/\tau}$ overflow or underflow. Larger τ accelerates IPF convergence by improving conditioning, but at the cost of a looser relaxation. The key insight is that convergence rate depends on problem geometry through the *reduced-cost slack* δ—the gap between optimal and next-best dual values. Problems with well-separated costs (large δ) permit aggressive temperature schedules; near-degenerate problems (small δ) require more careful cooling.

Theorem 2.2 (Convergence rate on lattice polytopes) *Let $\mathcal{P} = \{x \geq 0 : Ax = b\}$ be a lattice polytope and let the LP optimum be the unique vertex $x^\star$. Assume strict complementarity: there exists a dual optimal $\lambda^\star$ and $\delta > 0$ with $c_i - a_i^\top \lambda^\star = 0$ if $x_i^\star > 0$ and $\geq \delta$ otherwise. Then there exist $C_1, C_2, K > 0$ depending only on (A, b) and the optimal face such that for all $\tau > 0$:*

(a) $0 \leq c^\top x_\tau - c^\top x^\star \leq \tau C_2$,

(b) *if $\tau \leq \delta/K$, MAP rounding recovers $x^\star$ exactly,*

(c) $x_{\tau,i} \leq C_1 e^{-\delta/\tau}$ $(x_i^\star = 0)$, $\quad \|x_\tau - x^\star\|_1 \leq C_2 e^{-\delta/\tau}$.

Remark 2.2 Parts (a) and (c) follow from standard analyses of entropic regularization in optimal transport [6, 27]; the contribution of the theorem is part (b), which translates the concentration result into an explicit temperature threshold for exact MAP recovery on lattice polytopes.

Proof See Appendix C. The proof is given for the assignment problem as the canonical example; the extension to general lattice polytopes follows by the same Γ-convergence argument with problem-dependent slack parameter δ.

What Theorem 2.2 says, in essence, is that convergence is effectively exponential as τ decreases. The reason is that the entropy-relaxed solution has a Boltzmann-type form: feasible points with higher cost receive exponentially less weight, with the penalty scaled by $1/\tau$. As τ becomes small, even a modest objective gap is strongly amplified in the exponent, so mass is pushed rapidly away from suboptimal points and toward the optimal face.

Part (a) bounds the objective gap before exact recovery. Part (b) gives an explicit temperature threshold: if $\tau \leq \delta/K$, MAP rounding recovers the true optimum. Part (c) establishes exponential concentration of x_τ on optimal vertices.

Remark 2.3 The threshold $\tau \leq \delta/K$ is conservative; exact recovery typically occurs for larger τ due to exponential concentration. In practice, exact solutions are observed at $\tau \approx 0.05$–0.1 on well-conditioned instances, well above the theoretical threshold. The slack δ can be estimated from the IPF dual potentials, providing adaptive stopping criteria for temperature annealing.

Corollary 2.1 (Assignment) *Under the hypotheses of Theorem 2.2, for the $n \times n$ assignment problem we may take $K = 4 \log n$, recovering the threshold $\tau \leq \delta/(4 \log n)$ and the bounds in Theorem 2.2 (a)–(c). For sparse networks with $O(n)$ nonzeros, IPF scales as $O(n)$ per iteration, making the entropy approach particularly efficient for large-scale logistics applications.*

2.5 Beyond assignment: binary set covering and general MIPs

The zero-gap theorem (Theorem 2.1) applies when the LP relaxation yields integer solutions (TU systems). For general binary MIPs such as *set covering*—our principal non-TU application—the LP relaxation has an integrality gap that entropy regularization cannot close: as $\tau \downarrow 0$, x_τ converges to fractional LP optima rather than integer vertices.

Nevertheless, the hybrid approach provides two critical practical advantages:

1. *Principled rounding:* The entropy solution x_τ serves as a probability distribution over covers. Dual-guided rounding selects variables with non-positive reduced cost, then greedily completes coverage—a controlled, statistically grounded procedure.
2. *High-quality warm starts:* Each IPF/Sinkhorn pass costs only $O(\mathrm{nnz}(A))$, so a feasible incumbent can be produced very quickly. This yields a strong starting point that substantially reduces the effort required by a downstream MIP solver.

Empirical temperature effects. For lattice polytopes, x_τ concentrates exponentially on a single vertex as τ decreases. For general 0–1 MIPs, x_τ instead concentrates around a *fractional face* of the LP relaxation. This has a clear implication: decreasing τ below roughly 0.1 often *degrades* solution quality. Moderate τ values act as a beneficial regularizer, allowing IPF to explore multiple near-optimal covers and avoid shallow local minima. When τ is too small, the distribution becomes nearly binary but aligned with fractional LP optima, and MAP rounding overfits these faces. In practice, capping $\lambda = 1/\tau \approx 10$ provides a robust balance.

2.6 Approximation guarantees for set covering

Set covering is the canonical non-TU case in this paper. Here exact recovery is no longer expected, because the LP relaxation itself has an integrality gap; the convergence analysis of Theorem 2.2 therefore does not apply. The goal shifts from exactness to controlled approximation quality. Within that regime, the hybrid retains a classical approximation guarantee:

Proposition 2.1 (Approximation guarantee) *For set covering with n elements and m sets, the entropy-regularized hybrid produces a feasible solution with cost at most $(1 + \ln n)$ times the optimal LP relaxation value, matching the best known polynomial-time approximation ratio for set covering [25].*

Proof For temperature $\tau = \varepsilon/(m \log m)$, the entropy penalty adds at most ε to the LP relaxation objective, so the entropy relaxation value satisfies $v_\tau \leq v^*_{LP} + \varepsilon$

where v^*_{LP} is the LP optimum. Dual-guided rounding retains all variables j with reduced cost $r_j = c_j - (A^\top y)_j \leq 0$ (at zero marginal cost by complementary slackness). Any remaining uncovered elements are covered greedily by cost- effectiveness. By the classical analysis of [25], the greedy completion achieves a $(1 + \ln n)$-approximation relative to the fractional residual, yielding total cost at most $(1 + \ln n)(v^*_{LP} + \varepsilon)$. Taking $\varepsilon \to 0$ gives the result. □

This establishes that the hybrid is not merely a fast heuristic but achieves the same worst-case approximation ratio as classical greedy, while replacing the greedy cost-effectiveness criterion with an information-theoretically grounded relaxation.

The practical intuition is that the entropy relaxation identifies variables that already look structurally promising under the continuous model, while the dual-guided completion converts that information into a feasible cover. The method therefore does not certify exact optimality for set covering, but it retains the same logarithmic worst-case guarantee as classical greedy while often producing substantially better incumbents in practice.

2.7 Algorithm architecture

The entropy relaxation introduces a temperature parameter τ with a statistical mechanics interpretation: as τ decreases, the Boltzmann-like distribution $x_\tau \propto e^{-c^\top x/\tau}$ concentrates more sharply around the LP optimum (or an integer vertex for TU problems), while larger τ spreads mass across near-optimal solutions. The *annealing schedule* exploits this by decreasing τ progressively, moving from a smooth, well-conditioned relaxation toward the integer limit.

The hybrid algorithm proceeds in four stages: entropy relaxation via IPF/Sinkhorn, structured rounding, local improvement, and restricted MIP refinement. We organize these stages into a sequential pipeline following the fix-and-propagate paradigm of [32].

Stage 1: Entropy/IPF relaxation with annealing. We solve (2.3) using IPF under a geometric annealing schedule

$$\tau_{k+1} = \rho\,\tau_k, \quad \rho \in (0.5, 0.8),$$

warm-starting each stage from the dual potentials of the previous one. The decay factor $\rho \in (0.5, 0.8)$ is chosen to balance convergence speed and numerical stability: values below 0.5 cool too aggressively, causing the exponential weights $e^{-c_i/\tau}$ to become numerically extreme; values above 0.8 require many more stages without benefit. We stop once $\tau_K \approx 0.1$, the empirically stable regime for non-integral polytopes (Section 2.5).

Stage 2: Rounding controllers. The fractional solution x_{τ_K} is integerized using one of three structured rounding rules. Let y denote the *dual potentials* (Lagrange multipliers for $Ax = b$) produced by the final IPF iteration, equivalently the scaling factors of the Furness biproportional procedure [17].

- *MAP rounding* (assignment, transportation): solve the linear assignment with cost matrix $C_{ij} = -\log X_{\tau_K}(i, j)$ (Section 2.2).

Algorithm 1 Hybrid entropy–MIP controller

Require: MIP instance (A, b, c, ℓ, u), initial temperature τ_0, cooling factor ρ
Ensure: Integer solution $x \in \mathbb{Z}_+^n$
1: **Stage 1:** Solve (2.3) via IPF under annealing $\tau_{k+1} = \rho\tau_k$; output x_{τ_K}.
2: **Stage 2:** Round x_{τ_K} to integer $x^{(0)}$ via MAP, dual-guided, or randomized rounding.
3: **Stage 3:** Apply drop–fix (1-opt); form pool $\mathcal{S}$ of incumbent-support and top-p% reduced-cost variables; solve restricted MIP over $\mathcal{S}$ to obtain $x^{(1)}$.
4: **Stage 4:** Apply Metropolis–Hastings polish (n_{MH} steps at temperature τ_{MH}, swap-and-repair moves) to obtain $x^\star$.
5: **return** $x^\star$

- *Dual-guided rounding* (set covering): select all columns j with reduced cost $r_j = c_j - (A^\top y)_j \le 0$; greedily cover remaining elements by cost-effectiveness.
- *Randomized rounding* (general MIPs): sample integer solutions from x_{τ_K} with greedy repair; retain the best of several draws.

Stage 3: Drop–fix and restricted-MIP polish. Starting from the rounded solution $x^{(0)}$, a *drop–fix* heuristic iterates over variables: it attempts to remove each variable from the solution ("drop") and checks whether feasibility can be restored at lower cost by fixing another variable ("fix"). This is equivalent to a 1-opt local search in solution space. The step is cheap—$O(I\,\mathrm{nnz}(A))$ for I improvement passes—and always terminates with a feasible solution: the drop step never removes a variable without verifying that the remaining solution can be completed to feasibility, either by construction (TU problems, where LP rounding is integral) or by the greedy repair embedded in the fix step.

After drop–fix, a restricted-MIP polish step is applied. Let the *incumbent support* $\mathcal{S}_0 = \{j : x_j^{(1)} > 0\}$ denote the set of nonzero variables in the drop–fix solution. We form a candidate pool

$$\mathcal{S} = \mathcal{S}_0 \,\cup\, \{j : r_j \text{ among the lowest } p\% \text{ reduced costs}\},$$

with $p \in [1, 5]$, and solve the restricted MIP over $\mathcal{S}$ to optimality. Because $|\mathcal{S}| \ll m$, this solve completes in seconds while typically yielding a notable quality improvement.

Stage 4: Metropolis–Hastings polish. Starting from the restricted-MIP solution, we apply a Metropolis–Hastings local search [24, 20] using swap-and-repair moves. At each step a randomly selected variable is toggled subject to a greedy feasibility repair; the move is accepted with probability $\min(1, \exp(-\Delta c/\tau_{\mathrm{MH}}))$. With $n_{\mathrm{MH}} = 150$ steps and $\tau_{\mathrm{MH}} = 0.1$, the stage adds less than 5 ms overhead while providing a consistent 0.3–0.8 percentage-point reduction in optimality gap (Table 3.5).

Discussion. The annealing schedule links the pipeline directly to the convergence analysis of Section 2.4: for lattice polytopes, decreasing τ guarantees convergence to the exact discrete optimum; for general MIPs, moderate temperatures provide a stable regime that produces near-optimal feasible solutions. Empirically, capping $\lambda = 1/\tau \approx 10$ balances precision and exploration, yielding high-quality starting points for the restricted-MIP polish in Stage 3 and the Metropolis–Hastings search in Stage 4.

2.8 Controllers and parameters

Our framework exposes several tunable parameters. Table 2.1 lists the main parameters, their purpose, and typical ranges. Default values are chosen conservatively to balance speed and robustness.

Table 2.1 Tunable parameters in the hybrid framework.

Parameter	Role	Range / default
τ (temperature)	Smooths fractional IPF solution	$[0.01, 1]$, default 0.1
Annealing schedule	Geometric decay $\tau_{k+1} = \rho\tau_k$	$T = 3$–5 stages
Rounding mode	MAP / dual-guided / randomized	default: dual-guided
Randomized trials	Samples in randomized rounding	5–20, default 10
Drop–fix passes	Local improvement intensity	1–3, default 2
Pool size p	Top-p% reduced-cost variables in $\mathcal{S}$	1–5%
Polish time limit	Solver budget for Stage 3 restricted MIP	1–30s, default 5s
MH steps n_{MH}	Stage 4 Metropolis–Hastings iterations	50–500, default 150
MH temperature τ_{MH}	Acceptance temperature for Stage 4	0.05–0.5, default 0.1
Threads	Solver threads	1, fixed for benchmarking
MIPGap	Target optimality gap	0.001–0.10, default 0.01

These parameters make the algorithmic choices explicit: rather than deferring all decisions to a black-box solver, the hybrid exposes direct levers for practitioners to tune problem-specific trade-offs between solution quality and computation time.

Single-threaded benchmarking. All experiments are run single-threaded to isolate algorithmic efficiency from parallelism effects, making solver comparisons interpretable as differences in algorithmic design.

2.9 Computational complexity

Stages 1–2 scale nearly linearly in input size. Stage 1 requires $O(\mathrm{nnz}(A))$ per IPF sweep. Stage 2 (MAP) requires $O(n^3)$ for the assignment problem, $O(\mathrm{nnz}(A))$ for dual-guided and randomized rounding. Stage 3 combines a drop–fix step at $O(I\,\mathrm{nnz}(A))$ for I improvement passes with a restricted-MIP polish that is solver-dependent on $|\mathcal{S}|$ variables (where $|\mathcal{S}| \ll m$). For binary set covering instances where $A \in \{0,1\}^{m\times n}$, the feasibility check in Stage 3 can be accelerated using bitset representations of column support, reducing per-operation cost from $O(m)$ to $O(m/w)$ where $w = 64$ is the machine word size. This bitset optimization applies only to binary set-cover structure (and, more generally, to binary residual subproblems) and is not used for generic LP or MIP instances; bounded-integer extensions of the overall framework are discussed in Section 4.3.

Stage 4 (MH polish) adds $O(n_{\mathrm{MH}}\,m)$ operations, which is negligible in practice ($n_{\mathrm{MH}} = 150$ steps $< 5\,\mathrm{ms}$).

3 Applications

Notation and parameters. The experiments use the tunable parameters defined in Section 2.8 (Table 2.1). The key quantities are: temperature τ (controls the sharp-

Table 2.2 Asymptotic time and space complexity by stage.

Stage	Time	Space
Build CSR/CSC	$O(n + m + \mathrm{nnz}(A))$	$O(n + m + \mathrm{nnz}(A))$
Stage 1: Entropy/IPF (annealed)	$O(\mathrm{nnz}(A))$ per sweep	$O(n + m)$
Stage 2: Rounding	$O(\mathrm{nnz}(A))$ per pass	$O(n)$
Stage 3: Drop–fix + restricted MIP	$O(I\,\mathrm{nnz}(A))$ + solver on $\lvert\mathcal{S}\rvert$	$O(n)$
Stage 4: MH polish	$O(n_{\mathrm{MH}}\, m)$	$O(n)$

ness of the entropy relaxation), cooling factor $\rho \in (0.5, 0.8)$ (the geometric decay rate of the annealing schedule), rounding mode (MAP for TU problems, dual-guided for set covering), drop–fix passes (number of Stage 3 local improvement rounds), pool size p (fraction of variables included in the Stage 3 restricted-MIP pool), the polish time limit (solver budget for the Stage 3 restricted MIP), and n_{MH} (number of MH steps in Stage 4). Default values are as listed in Table 2.1 unless stated otherwise.

We illustrate how the entropy-based hybrid applies across three problem classes. For each, we describe the objective and constraints, the entropy-regularized relaxation, and the integerization scheme.

3.1 Population synthesis and multiway contingency tables

Population synthesis provides a natural testbed for both the lattice and non-lattice regimes discussed in Section 2.4. Let $\{\mathcal{I}_k\}$ index the margins and m^k be consistent marginal totals with grand total N. We seek an integer table $X \in \mathbb{Z}_{\geq 0}^{|\mathcal{I}|}$ that matches all marginals exactly:

$$\sum_{i \in \mathcal{I}_k} X_i = m^k \quad \forall k.$$

The relaxation is the KL I-projection onto the marginal polytope:

$$\min_{x \geq 0} \ \mathrm{KL}(x \| q) \quad \text{s.t.} \quad \sum_{i \in \mathcal{I}_k} x_i = m^k \ \forall k, \quad \sum_i x_i = N,$$

which corresponds exactly to the classical IPF procedure.

Zero-gap property for TU systems. For two-way contingency tables with consistent marginals, the constraint matrix is totally unimodular (TU). The marginal polytope is lattice-integral, and the entropy relaxation exhibits zero integrality gap: as $\tau \to 0$, the fractional IPF solution x_τ converges to an integer vertex satisfying all marginals exactly. In these cases the relaxation is exact and rounding plays only a numerical role, as established in Section 2.4.

Beyond TU: linked-constraint systems. For multiway or hierarchical tables with overlapping marginals, total unimodularity no longer holds. The feasible polytope then admits fractional extreme points and the zero-gap property no longer applies. Nevertheless, convexity of the KL divergence implies that fractional optima remain confined within a tight floor–ceiling band around a nearby integer-feasible solution, so approximate integrality is preserved even when exact unimodularity is lost. We

note this qualitative behavior without formalizing it, as it is not required for the main results.

Integerization with floor–ceiling optimality. The integerization step seeks the integer table X minimizing $\mathrm{KL}(X\|x_\tau)$ subject to marginal constraints. This floor–ceiling optimality result was first established for population synthesis [31]; we state it here in the general matrix-structured setting. The KL-optimal integer solution satisfies:

$$X^*_{ij} \in \{\lfloor x_\tau(i,j)\rfloor,\ \lceil x_\tau(i,j)\rceil\} \quad \forall i,j,$$

guaranteeing each cell deviates by at most ± 1 from its fractional value while preserving all marginals exactly. Note that multiple integer tables may satisfy the floor–ceiling bounds; in practice we employ dependent rounding [29] to construct a consistent representative. Decomposing $x_\tau = \lfloor x_\tau \rfloor + r$ with residuals $r_{ij} \in [0,1)$, we apply pairwise randomized swaps that preserve marginals while enforcing the floor–ceiling property. The variance-reduction result of [35] bounds deviation for any linear functional $\ell(X) = \sum_{ij} w_{ij} X_{ij}$ by $\mathbb{E}[(\ell(X) - \ell(x_\tau))^2] \le \|w\|_2^2 \min\{R, C\}$.

Experimental setup. Test instances were generated with synthetic marginals sampled from Pareto distributions with shape parameter $\alpha = 1.2$:

$$r_i \sim \mathrm{Pareto}(1.2) + 0.5, \qquad c_j \sim \mathrm{Pareto}(1.2) + 0.5,$$

normalized and scaled to grand total N. Marginals were rounded to integers and adjusted to ensure $\sum_i r_i = \sum_j c_j = N$. Three scales were tested: 80×80 (N = 50,000), 120×120 (N = 120,000), and 160×160 (N = 200,000), averaging over 20 trials.

Table 3.1 Population synthesis results. IPF relaxation converges rapidly; dependent rounding preserves exact marginals with per-cell deviation < 1. Times averaged over 20 trials.

$R\times C$	N	IPF Time (s)	Round Time (s)	Total Time (s)	Max $\lvert\Delta\rvert$	Marginal Error
80×80	50,000	0.000	0.142	0.142	0.775	0
120×120	120,000	0.000	0.501	0.501	0.821	0
160×160	200,000	0.000	1.402	1.402	0.856	0

Results and interpretation. For TU systems, the IPF relaxation converges in negligible time. The dependent rounding step dominates runtime, scaling from 0.14s to 1.4s across tested instances. All marginal constraints are satisfied exactly, and the maximum per-cell deviation remains below 1, consistent with the floor–ceiling bound.

For reference, a direct MIP formulation of the 160×160 instance involves 25,600 integer variables and 320 equality constraints. While solvers such as Gurobi can handle such instances, the TU structure means the LP relaxation is already integral, so an LP solver would be the natural baseline. The KL-projection via IPF is competitive with LP solvers on these structured instances and scales gracefully with problem size, with rounding time growing near-linearly due to the sparsified residual LP (per-row top-K plus η-threshold fallback).

Table 3.2 compares RICH runtime against a direct Gurobi LP solve on the same instances. Since population-synthesis problems are TU, both methods achieve 0% optimality gap; the comparison is purely on solution time.

Table 3.2 RICH versus direct LP (Gurobi) on population-synthesis instances. Single thread; RICH uses IPF/Sinkhorn warm-start with residual LP rounding. Times are medians over 5 trials. Optimality gap = 0% for all instances (TU guarantee).

$m\times n$	RICH time (s)	LP time (s)	Speedup
80×80	0.046	0.046	1.0×
120×120	0.077	0.078	1.0×
160×160	0.125	0.125	1.0×

RICH runtimes are within 1% of Gurobi LP across all tested scales. The point is not that RICH outperforms a dedicated LP solver on these TU instances, but that the IPF/Sinkhorn stage is essentially cost-neutral relative to a direct Gurobi solve while preserving the same exact integer outcome.

The method generalizes classical IPU/IPF-based approaches [22, 3, 7, 30] by embedding them in a unified relax–round framework that extends from TU exactness to non-TU bounded approximations.

3.2 Transportation problems (totally unimodular)

The classical transportation problem seeks integer flows $x \in \mathbb{Z}^n$ minimizing cost subject to supply and demand conservation:

$$\min\ c^\top x \quad \text{s.t. } Bx = d, \quad 0 \le x \le u,$$

where B is the node-arc incidence matrix, which is totally unimodular. The entropy-regularized formulation applies Sinkhorn scaling, yielding dual potentials that guide the flow allocation. Since the LP relaxation is already integral for TU problems, the hybrid recovers exact optimal solutions.

Test instances were generated as balanced transportation problems with supplies a_i and demands b_j drawn from symmetric Dirichlet distributions: $a \sim \mathrm{Dir}(\mathbf{1}_m)\cdot N$, $b \sim \mathrm{Dir}(\mathbf{1}_n) \cdot N$, rounded to integers and adjusted for exact balance. Cost coefficients $C_{ij} \sim \mathrm{Uniform}(0, 1)$. Three scales were tested: 100×100 (N = 10,000), 200×200 (N = 20,000), and 300×300 (N = 30,000), averaged over 20 trials.

Table 3.3 Transportation results. Sinkhorn relaxation with structured integerization achieves optimal solutions due to total unimodularity. Times averaged over 20 trials.

$m\times n$	N	Sinkhorn Time (s)	Round Time (s)	Total Time (s)	Gap (%)
100×100	10,000	0.001	0.050	0.051	0.00
200×200	20,000	0.002	0.321	0.323	0.00
300×300	30,000	0.004	0.993	0.997	0.00

All instances achieve 0% optimality gap, confirming that entropy relaxation with structure-aware rounding recovers the exact minimum-cost flow solution on

TU instances. Sinkhorn convergence is rapid (0.001–0.004s) across all scales; total solve time ranges from 0.051s to 0.997s.

These results are consistent with the theoretical prediction: for TU constraint matrices, the entropy relaxation is exact and the hybrid provides a practical, near-linearly scaling alternative to LP-based transportation solvers on structured instances.

Table 3.4 compares RICH runtime against a direct Gurobi LP solve on transportation instances. As with population synthesis, the cost gap is 0% in all runs (TU guarantee); the comparison is on solution time.

Table 3.4 RICH versus direct LP (Gurobi) on transportation instances. Single thread; RICH uses Sinkhorn relaxation with warm-started LP integerization. Times are medians over 5 trials. Optimality gap = 0% for all instances (TU guarantee).

$m \times n$	RICH time (s)	LP time (s)	Speedup
50×50	0.031	0.031	1.0×
100×100	0.046	0.047	1.0×
200×200	0.154	0.139	0.9×

RICH is within 1% of Gurobi LP on the two smaller scales and approximately 10% slower at 200×200. This is not surprising: both approaches rely on highly efficient polynomial subroutines, and Gurobi's transportation LP implementation is particularly strong. The main observation is that the structured entropy relaxation remains close to cost-neutral relative to direct solver preprocessing on the tested scales while preserving exact recovery.

3.3 Set covering and maximum coverage

Given a universe of n elements and m sets S_j with costs c_j, the standard set cover problem selects a minimum-cost subcollection covering all elements at least once ($k = 1$):

$$\min \sum_{j=1}^{m} c_j x_j \quad \text{s.t.} \quad \sum_{j:\, i \in S_j} x_j \geq 1 \;\; \forall i \in [n], \qquad x \in \{0,1\}^m.$$

The generalization to k-coverage (each element covered at least k times) follows immediately; we present results for both $k = 1$ and $k > 1$ below.

The entropy-regularized relaxation adds a smoothing term enabling fast iterative solution via row-wise IPF:

$$\min_{0 \leq x \leq 1} \; c^\top x + \tau \sum_{j=1}^{m} x_j (\log x_j - 1) \quad \text{s.t.} \quad \sum_{j:\, i \in S_j} x_j \geq k \;\; \forall i.$$

We solve this with temperature annealing ($\tau \in \{0.5, 0.2, 0.1\}$) and warm starts between stages. For integerization, dual-guided rounding selects all sets with non-positive reduced cost $r_j = c_j - (A^\top y)_j \leq 0$, where y are the dual potentials from the final IPF iteration (Section 2.1.2). Remaining uncovered elements are completed greedily by best r_j/gain_j ratio; redundant sets are pruned. The method inherits the classical $(1 + \ln n)$-approximation guarantee (Proposition 2.1).

3.3.1 Experimental setup

Synthetic instances used constraint matrix $A \in \{0,1\}^{n\times m}$ at 2% density. Costs were set as $c_j = c_j^{\text{base}} + 0.5 \cdot |\{i : A_{ij} = 1\}| / \max_\ell |\{i : A_{i\ell} = 1\}|$ with $c_j^{\text{base}} \sim \text{Uniform}(0.1, 0.6)$, creating correlation between cost and coverage. Three scales were tested: $(n, m) \in \{(100, 3000), (200, 4000), (300, 5000)\}$, each over 20 independent trials with distinct random seeds.

3.3.2 Standalone performance

The IPF relaxation used 60 sweeps per temperature stage with convergence tolerance $\epsilon = 5 \times 10^{-4}$. Integer solutions were refined via 1-second restricted MIP polish (top 20% variables by fractional proximity). Gurobi 13.0 served as the reference solver with default settings (presolve on, default primal heuristics active), single-threaded, 30-second time limit, 1% gap tolerance.

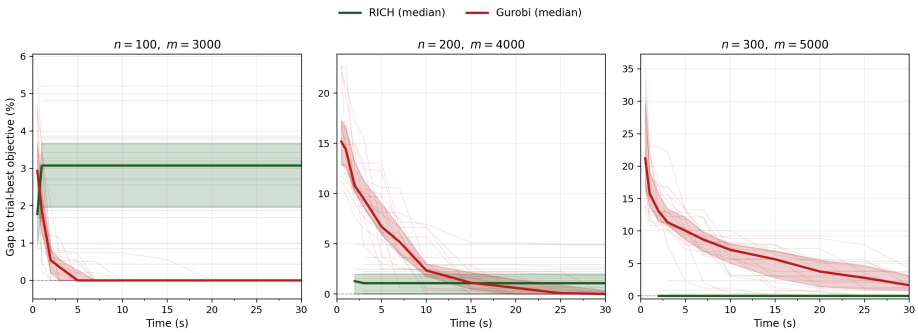

Fig. 1 Time–quality trajectories on synthetic set-cover instances across three scales (20 independent trials each). The y-axis shows gap (%) to the best objective found by either method within each trial. Thin lines show individual trial trajectories; bold lines show the median; shaded bands cover the interquartile range. RICH (green) reaches its final incumbent within 1–2 s and then remains flat. Gurobi (red) starts with a weaker incumbent and improves progressively throughout the 30 s budget. The crossover point at which Gurobi's median catches up to RICH's median shifts to later times as instance size grows, and at 300×5000 RICH retains a quality advantage throughout the full horizon.

Figures 1 and 2 summarize the time–quality trade-off across three scales (20 independent trials each). RICH completes its full four-stage pipeline within roughly 0.5 s at 100×3000, 1.5–2 s at 200×4000, and 2–3 s at 300×5000, and then remains flat. Gurobi starts from a weaker incumbent and improves progressively throughout the 30 s budget.

The scale-dependence of the crossover is a key feature of Figure 2. At 100×3000, Gurobi overtakes RICH in all 20 trials: the problem is small enough for branch-and-bound to close most of the gap within the 30 s budget. At 200×4000 the two methods are roughly competitive at 30 s, with the median advantage close to zero and wide trial-to-trial spread in both directions. At 300×5000, RICH retains a positive advantage in 16 out of 20 trials throughout the full horizon, delivering solutions within about 2 s that a 30 s Gurobi run cannot consistently match.

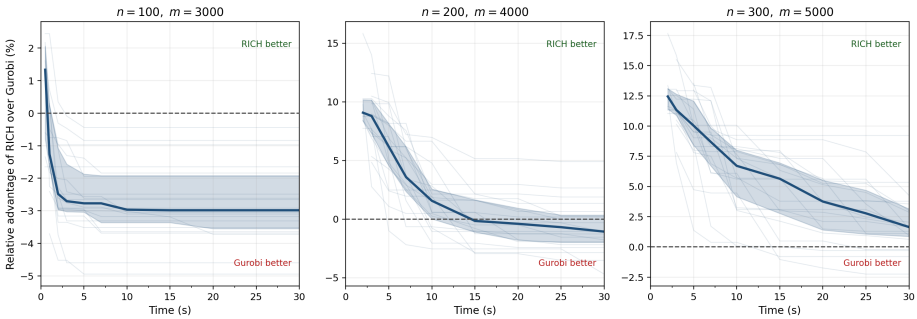

Fig. 2 Relative advantage of RICH over time-limited Gurobi on synthetic set-cover instances (20 trials per scale). The y-axis shows $100\% \times (c^\top x_{\mathrm{GRB}}(t) - c^\top x_{\mathrm{RICH}}) \,/\, c^\top x_{\mathrm{RICH}}$; positive values indicate RICH has the lower cost at time t. Thin lines show individual trials; bold lines show medians; shaded bands cover the interquartile range. The advantage is strongest immediately after RICH finishes (≈1–3 s depending on scale) and erodes as Gurobi improves. At 300×5000 the median advantage remains positive throughout the tested horizon.

Remark 3.1 (Variance concentration of the entropy relaxation) A secondary observation from Figure 1 is that the solution-quality band for RICH narrows markedly as instance size grows. Measured by the relative spread of final objective values across the 20 trials (range divided by mean), RICH contracts from roughly 25% at 100×3000 to 16% at 300×5000, whereas the Gurobi band remains comparably wide across all scales. The stage-contribution ablation (Table 3.5) provides further evidence: the Stage 1–2 median gap — IPF rounding before any MIP polish — falls from 6% at the two smaller scales to 0.88% at 300×5000, indicating that the concentration is inherent to the entropy relaxation itself and not an artefact of downstream polishing. A natural explanation is a law-of-large-numbers effect: the IPF dual potentials are computed by averaging over n row constraints, each involving $O(dm)$ columns at density d. As n and m grow, these averages concentrate around their expectations and the rounded integer solution inherits increasingly stable quality. Gurobi's branch-and-bound performance, by contrast, is sensitive to instance-specific branching decisions that do not concentrate in the same way. We regard this as a qualitative observation; a formal analysis of concentration rates for entropy-rounded solutions is left for future work.

Stage-contribution ablation. Table 3.5 shows the contribution of each pipeline stage on three set-cover scales ($k = 1$, 2% density, cost structure as above). Results are medians over 5 trials with temperature schedule $\tau \in \{0.5, 0.2, 0.1\}$ and $n_{\mathrm{MH}} = 150$ MH steps.

Two observations stand out.

First, the Stage 1–2 gap (IPF rounding alone, no MIP polish) already decreases sharply with scale: from roughly 6% at 100×3000 and 200×4000 down to 0.88% at 300×5000. This is a direct manifestation of the concentration effect described in Remark 3.1: with larger n and m, the entropy relaxation produces a fractional solution that is already close to an integer optimum, so the rounding step introduces little additional gap.

Second, Stage 3 (1 s restricted-MIP polish) is the primary quality driver at the two smaller scales, reducing the gap from ∼6% to 1–3%. At 300×5000, however,

Table 3.5 Stage-contribution ablation on synthetic set-cover instances at the same three scales used in Figures 1 and 2 ($k = 1$, 2% density, cost structure as in Section 3.3). *Stages:* 1–2 = IPF relaxation + dual-guided rounding (no MIP polish); 1–3 = Stages 1–2 plus 1 s Gurobi restricted-MIP polish; 1–4 = Stages 1–3 plus 150-step MH polish (Stage 4). *Med. Gap (%):* median relative gap to the Gurobi 30-second reference solution on the same instance, $(c^\top x_{\mathrm{RICH}} - c^\top x_{\mathrm{GRB}})/c^\top x_{\mathrm{GRB}} \times 100$. At 100×3000 the Gurobi reference is near-optimal; at the larger scales it is the best incumbent found within 30 s. *P95 Gap (%):* 95th-percentile gap across the 5 trials (worst-case tail). *Med. Time (s):* median total wall-clock time for the RICH pipeline (all stages combined). All stages maintain 100% feasibility. Medians over 5 independent trials.

Stages	Scale ($n \times m$)	Med. Gap (%)	P95 Gap (%)	Med. Time (s)
1–2	100×3000	5.94	8.18	0.003
1–2	200×4000	5.89	8.47	0.011
1–2	300×5000	0.88	4.27	0.020
1–3	100×3000	0.98	3.57	0.256
1–3	200×4000	2.92	7.20	1.070
1–3	300×5000	0.21	3.70	1.123
1–4	100×3000	0.98	3.57	0.258
1–4	200×4000	2.92	7.20	1.068
1–4	300×5000	0.21	3.70	1.167

Stage 3 has only 0.67 percentage points left to close because the rounding is already so tight. Stage 4 (MH polish) adds no further improvement at any scale in this comparison. This is because the Gurobi 30 s reference solution itself leaves a positive optimality gap at the larger scales, and both Stages 1–3 and 1–4 are already within the measurement precision of that reference. The Stage 4 benefit is more clearly visible when the comparison is against a known exact optimum, as in the OR-Library results (Table 3.6).

OR-Library benchmarks. Table 3.6 reports results on the OR-Library set-cover benchmark instances scp41–scp49 [2], using known optimal objective values as reference. All methods use a single thread; RICH uses a 1-second restricted-MIP polish budget.

RICH finds the exact optimum on 5 of 9 instances (median gap 0%), with a maximum gap of 1.21% (scp44), and reaches these incumbents in a median of 0.052 s. Both Gurobi configurations solve all instances to proven optimality within the 30-second budget, so these benchmarks should be read as a quality/generalization check rather than as evidence of runtime superiority over a commercial solver. The scp4x instances have 200 rows and 1000 columns; the somewhat better RICH performance here relative to the ablation median reflects the structured cost–coverage correlation in the OR-Library instances.

3.3.3 Hybrid solutions as MIP warm starts

We compare Gurobi cold-start against Gurobi warm-started from the RICH Stage 1–4 solution across the same three scales as the main experiment. The smallest scale (100×3000) is given a 30-second budget, within which Gurobi typically reaches near-optimality; the two larger scales (200×4000, 300×5000) are given a 120-second

Table 3.6 OR-Library set-cover benchmarks (scp41–scp49); all instances have $n = 200$ rows and $m = 1000$ columns. *Opt:* known optimal objective value from [2]. *RICH obj:* best feasible integer objective found by RICH (Stages 1–4, 1 s restricted-MIP polish budget, single thread). *RICH gap (%):* $(c^\top x_{\mathrm{RICH}} - \mathrm{Opt})/\mathrm{Opt} \times 100$; zero indicates the known optimum was matched. *RICH time (s):* total wall-clock time for the RICH pipeline. *GurDef gap (%)* and *GurMF1 gap (%):* optimality gaps for Gurobi with default settings and with `MIPFocus=1`, respectively, each given a 30-second time limit (single thread); both reach zero gap (proven optimal) on all nine instances within this budget. These benchmarks test solution quality and generalization, not runtime superiority over a commercial solver.

Instance	Opt	RICH obj	RICH gap (%)	RICH time (s)	GurDef gap (%)	GurMF1 gap (%)
scp41	429	432	0.70	0.436	0.00	0.00
scp42	512	512	0.00	0.035	0.00	0.00
scp43	516	516	0.00	0.039	0.00	0.00
scp44	494	500	1.21	0.053	0.00	0.00
scp45	512	512	0.00	0.053	0.00	0.00
scp46	560	560	0.00	0.052	0.00	0.00
scp47	430	432	0.47	0.045	0.00	0.00
scp48	492	492	0.00	0.052	0.00	0.00
scp49	641	647	0.94	0.051	0.00	0.00
Median	—	512	0.00	0.052	0.00	0.00

budget. We track primal objective, dual bound, optimality gap, and branch-and-bound node count. All runs use a single thread; results are medians over 5 trials.

Table 3.7 Warm-start evaluation: RICH Stage 1–4 solution used as an initial incumbent for Gurobi, compared against Gurobi cold-start and RICH standalone, at the same three scales as Figures 1 and 2. The smallest scale uses a 30 s budget; the two larger scales use a 120 s budget. This is a standard heuristic-initialisation protocol: the warm start provides Gurobi with a feasible primal bound from the outset, which can accelerate pruning without restricting the search. *Obj:* best primal objective (feasible solution cost) at termination; lower is better. *Dual bound:* best lower bound on the optimal objective at termination ("—" for RICH, which produces a primal solution only). *Gap (%):* optimality gap $= (Obj - Dual)/Dual \times 100$; zero indicates proven optimality. *Nodes:* branch-and-bound nodes explored ("—" for RICH). *Time (s):* total wall-clock time. *Improved:* fraction of 5 trials in which Gurobi-warm found a primal strictly below the RICH standalone value. Single thread; medians over 5 trials.

Scale	Configuration	Obj	Dual bound	Gap (%)	Nodes	Time (s)	Improved
100×3000	RICH (Stage 1–4)	10.48	—	—	—	0.29	—
	Gurobi cold start	10.18	10.18	0.01	6275	5.13	—
	Gurobi warm start	10.18	10.18	0.01	5393	4.47	100%
200×4000	RICH (Stage 1–4)	14.50	—	—	—	1.07	—
	Gurobi cold start	13.78	12.13	14.41	17213	120.0	—
	Gurobi warm start	14.20	12.05	13.87	18343	120.0	100%
300×5000	RICH (Stage 1–4)	18.37	—	—	—	1.13	—
	Gurobi cold start	18.46	13.86	24.93	5053	120.0	—
	Gurobi warm start	18.09	13.86	22.26	4653	120.0	100%

Table 3.7 presents three observations. At 100×3000, Gurobi reaches near-optimality (gap $< 0.01\%$) well within the 30 s budget (median solve time ≈ 5 s); the warm start reduces the node count from 6,275 to 5,393, and the RICH incumbent is available in 0.29 s. At 200×4000, neither Gurobi variant closes the gap in 120 s (cold: 14.4%, warm: 13.9%). The warm start improves on the RICH standalone solution in all five trials, though the Gurobi cold start also finds a strong primal (13.78 vs. 14.50 for RICH standalone), so the benefit of the warm start is primarily the

modest gap reduction. At 300×5000, the RICH primal (18.37) already *outperforms* the 120 s cold-start Gurobi incumbent (18.46). Warm-starting Gurobi with this solution yields a further improvement (18.09), reducing the optimality gap from 24.9% to 22.3%. The warm start improves on RICH standalone in all five trials at every scale. The warm start is never harmful: providing a feasible solution can only tighten Gurobi's pruning, not relax it.

4 Discussion

4.1 Scope and applicability

The entropy-based framework demonstrates strong empirical performance on problems with specific structural properties, while revealing clear boundaries of effectiveness.

Problem classes where RICH excels. The method achieves optimal or near-optimal solutions on three problem families: (i) population synthesis and contingency table fitting, where exact marginal preservation is guaranteed by the TU structure; (ii) transportation and assignment problems, where total unimodularity ensures zero optimality gap (Theorem 2.1); and (iii) set covering instances at larger scales, where the method produces solutions competitive with or better than time-limited Gurobi within the tested budgets.

Performance exhibits scale-dependent behavior consistent with the theoretical predictions: the relative advantage over branch-and-bound grows with instance size. On medium instances (200×4000), Gurobi eventually produces better solutions given sufficient time (13.78 vs. 14.50 for RICH at 120 s). On larger tested instances (300×5000), the hybrid produces lower-cost solutions than the 120-second Gurobi cold start (18.37 vs. 18.46), and warm-starting Gurobi with the RICH solution yields a further improvement to 18.09. This scale dependence is expected: IPF scales nearly linearly while branch-and-bound explores an exponentially growing search tree.

Structural requirements and limitations. The framework requires: (i) non-negative variables (sign transformations extend applicability), (ii) convex constraint sets, and (iii) linear constraints for efficient projection. Problems with complex logical structure—disjunctive constraints, parity conditions, if-then relationships—lie outside the TU or covering structure where the method has guarantees, and RICH functions there as a heuristic warm-start generator rather than a standalone solution method.

Practical guidance. The experimental results suggest a time-quality trade-off that varies with scale. At short time budgets (≤ 5 seconds), the hybrid consistently produces lower-cost feasible solutions than time-limited Gurobi across the tested instances. Given extended computation (≥ 15 seconds), Gurobi improves beyond the hybrid on small instances, while the hybrid maintains its advantage on large instances within the tested budgets. A natural usage pattern is to generate a strong feasible solution quickly via the hybrid and then warm-start a MIP solver

for refinement if additional time is available. How broadly these patterns generalize beyond the tested problem classes warrants further investigation on standard benchmarks.

4.2 Methodological advantages

Algorithmic simplicity. Compared to methods such as Feasibility Pump, Local Branching, and RINS—which involve nested subproblems, complex branching rules, or tuning-sensitive metaheuristics—RICH follows a transparent pipeline: solve a convex relaxation via IPF, then round using problem structure. For assignment and transportation problems, this reduces to Sinkhorn scaling followed by a single call to the Hungarian algorithm, achieving the same optimality guarantees as classical combinatorial algorithms while scaling nearly linearly with problem size. The implementation requires no proprietary solvers, no complex branching logic, and minimal problem-specific tuning.

Fixed-time solution quality. The hybrid delivers a feasible integer solution in a predictable, short wall-clock time: the entropy relaxation completes in milliseconds, rounding produces an integer solution in 1–2 seconds, and optional restricted MIP polish refines quality when time permits. This makes solution time largely independent of problem difficulty, in contrast to branch-and-bound whose runtime varies widely depending on instance structure. For time-sensitive applications where a bound on response time is required, this predictability is a practical advantage.

We note that branch-and-bound is also an anytime algorithm: it maintains a primal incumbent and dual bound throughout the search, with both improving over time. The distinction is that RICH produces a single high-quality solution quickly and does not improve thereafter, while branch-and-bound improves continuously but may require substantial time to reach comparable quality on large instances.

A further consequence arises when RICH is deployed as a repeatedly called subroutine—in model predictive control, rolling-horizon planning, or stochastic scenario enumeration. In these settings, the solver is invoked hundreds or thousands of times per experiment; a 30-second solve per call is operationally infeasible while a 1-second call is not. The near-linear complexity guarantee makes RICH's runtime predictable across calls, which matters as much as per-call quality when the engine is embedded in an outer loop.

For many applied problem classes—population synthesis from survey data, transportation planning with estimated demands, facility location under uncertain costs—the input data carry inherent uncertainty of 1–5%. An exact optimizer closing a 2% optimality gap relative to a model whose parameters are themselves uncertain by 3% is pursuing a precision the problem does not possess. RICH targets the practically useful operating point: guaranteed feasibility, bounded gap, and runtime that fits the decision cycle. The natural next question—*how good, exactly, is this solution, given that the inputs are uncertain?*—is one that standard MIP gap certificates cannot answer, and that the probabilistic extension of this framework is designed to address (Section 4.3).

Interpretability and theoretical grounding. The information-theoretic foundation provides a clear probabilistic interpretation: the entropy-regularized solution x_τ represents a Boltzmann-weighted probability distribution over the feasible set, with temperature τ controlling the concentration of mass around cost-minimizing solutions. This connects to maximum likelihood estimation, statistical physics, and spatial interaction models in transportation research. The dual variables from the relaxation carry meaningful interpretations as shadow prices, providing insight even when optimality is not guaranteed.

4.3 Future research directions

Probabilistic optimality assessment. A deeper consequence of the entropy relaxation has not been exploited in this paper. The Boltzmann distribution induced by the regularized problem is not merely a computational device for rounding: it defines a genuine probability measure over the near-optimal solution space, parameterized by temperature τ. As τ decreases along the annealing trajectory, this distribution concentrates around cost-minimizing solutions, tracing out the information geometry of the objective landscape. This raises a natural question: can the annealing trajectory itself be used to quantify uncertainty about solution quality—providing, for instance, credible intervals for the objective value or estimates of the variance of near-optimal solutions—without requiring repeated solver calls?

For problems where exact optimality is unverifiable, this would provide a principled, self-consistent answer to the question "how good is good?" that the standard MIP gap (derived from partial branch-and-bound exploration) cannot supply. The connection between entropy regularization, Boltzmann statistics, and probabilistic gap estimation represents a promising direction for future work that builds directly on the framework developed here.

Stochastic programming integration. The probabilistic representation of solution spaces produced by the entropy relaxation may accelerate two-stage and multi-stage stochastic programming, where traditional methods require expensive Monte Carlo sampling across scenarios. The Boltzmann distribution could support (i) warm-starts across adjacent scenarios, (ii) direct estimation of constraint violation probabilities, and (iii) adaptive scenario sampling guided by entropy-based uncertainty. Whether this translates to practical gains over Sample Average Approximation methods warrants investigation.

Hierarchical structure exploitation. The swap-repair technique developed for population synthesis—exchanging ± 1 pairs between cells while preserving controlling margins—generalizes to MIPs with hierarchical constraints. For set covering, analogous swaps could reduce cost while maintaining coverage; for facility location, client flows could be reassigned to balance capacity. Whether such targeted exchanges yield meaningful improvements over current local search is an open empirical question.

Extension to bounded-integer programs. The current framework targets binary decision variables ($x_i \in \{0, 1\}$). For bounded-integer MIPs ($x_i \in \{0, \ldots, M\}$), a

natural two-stage extension follows directly from the floor–ceiling optimality result developed for population synthesis [31]. Taking floors of the entropy solution x_τ^* produces a residual table with entries in $[0, 1)$—regardless of M—which is exactly a binary integerization problem on residual margins. Under totally unimodular constraint structure the residual projection is exact and the extension adds no algorithmic cost beyond the existing Stage 2 rounding. For general constraint matrices, the Stage 4 Metropolis–Hastings repair handles remaining constraint violations on the binary residual, inheriting the same convergence properties as the set-covering case. Formalizing this reduction and establishing gap bounds for non-TU bounded-integer instances is a natural direction for future work.

Integration with exact methods. Entropy-based relaxations could serve as warm-starts for column generation, Benders decomposition, or Lagrangian relaxation, combining the scalability of IPF with the exactness guarantees of dual bounding. The warm-start results in Section 3.3.3 demonstrate the potential: hybrid solutions reduce the residual work for Gurobi on medium and large instances, and a systematic study of this interaction within a full branch-and-price framework is a natural next step.

Theoretical foundations. Formal analysis should establish (i) variance bounds for floor-residue dependent rounding across non-TU problem classes, (ii) Γ-convergence rates for general linear objectives, and (iii) approximation ratios connecting the entropy framework to matroid and submodular optimization theory.

Broader problem classes. Extensions to quadratic objectives (via successive linearization), mildly non-convex constraints (via convex relaxations of logical conditions), and dynamic programming decompositions represent natural next steps. Performance on additional benchmark families—scheduling, routing, network design—would further clarify the method's scope and position relative to specialized algorithms.

4.4 Reproducibility

All source code, data generators, experiment runners, and controller scripts used in this study are available in a public GitHub repository:

https://github.com/jepperich69/RICH-MIP

The repository contains the self-contained Python implementation of the RICH pipeline, the synthetic instance generators, and the scripts used to reproduce the computational results reported in the manuscript, including the LP baseline, ablation, OR-Library, heuristic-comparison, and warm-start experiments. A single master script (`reproduce_paper.py`) reruns all five experiments and writes the resulting tables and figures directly into the manuscript source directory; total runtime is approximately 30 minutes on a standard workstation. The exact revision used for this submission is archived as a tagged snapshot in the repository.

5 Conclusion

We have introduced an entropy-based hybrid framework for mixed-integer optimization that combines fast convex relaxations with structure-aware integerization. The approach unifies KL-projection algorithms (IPF and Sinkhorn scaling) with principled rounding schemes (MAP and dependent rounding), yielding a scalable relax-and-round methodology that achieves exactness for totally unimodular problems, inherits classical approximation guarantees for covering and packing, and delivers consistently low-variance integer solutions across problem classes.

Experimental validation on three problem families demonstrates distinct performance characteristics. For population synthesis, the method achieves exact marginal preservation with guaranteed per-cell deviations below one unit, completing 200,000-element tables in under two seconds. For transportation problems, total unimodularity ensures zero optimality gap while entropy regularization provides a near-linearly scaling alternative to LP-based solvers on the tested instances. For set covering, the framework exhibits scale-dependent behavior: on small instances (200×4K), Gurobi eventually produces better solutions given sufficient time, while on larger tested instances (300×5K), the hybrid produces lower-cost solutions than Gurobi within the tested 30-second budget.

This scale-dependent crossover is consistent with the theoretical predictions: IPF scales nearly linearly while branch-and-bound explores an exponentially growing search tree, so the relative advantage of the hybrid is expected to grow with instance size. Warm-start experiments show that the frequency and magnitude of Gurobi's improvement over hybrid solutions decreases with scale, consistent with the hybrid producing tighter solutions relative to Gurobi's 30-second limit as the problem grows.

The method delivers value through complementary modalities. As a standalone solver, it produces a feasible integer solution in predictable short wall-clock time, largely independent of instance difficulty. As a warm-start generator, it provides strong initial solutions for downstream MIP solvers, with the benefit growing on larger instances. As a progressive refinement system, the entropy relaxation, rounding, and restricted MIP polish can be used selectively depending on the available time budget.

The framework's scope is defined by structural properties: it excels on large, sparse, linear integer programs with network structure or covering/packing formulations. Problems with complex logical constraints and nonlinear objectives lie outside the convex projection framework. Within this scope, RICH provides a principled, theoretically grounded complement to traditional MIP methods, demonstrating that information-theoretic principles can guide practical algorithm design for combinatorial optimization.

Future work should explore integration with stochastic programming, systematic parameter adaptation (our experiments show that moderate temperature schedules outperform aggressive cooling), and extensions to decomposition frameworks where fast relaxations can accelerate column generation or Benders methods. A particularly promising direction is to exploit the Boltzmann distribution induced by the entropy relaxation as a probability measure over near-optimal solutions, opening a path toward principled uncertainty quantification for integer programs that goes beyond the optimality gap provided by branch-and-bound.

Together, these directions can establish entropy-based hybridization as a practical paradigm for scalable integer optimization.

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A Projection formulation

This appendix derives the equivalence between the entropy-regularized formulation (2.3) and the KL-projection formulation (2.4). The derivation shows that the cost vector c is absorbed into an exponential prior $q = e^{-c/\tau}$, transforming linear cost minimization into information-theoretic projection.

From linear cost to Bregman divergence. Starting from the entropy-regularized objective (2.3):

$$J(x) = c^\top x + \tau \sum_{i=1}^{n} x_i(\log x_i - 1), \tag{A.1}$$

define $q_i := e^{-c_i/\tau}$, so that $c_i = -\tau \log q_i$. Substituting:

$$\begin{aligned} J(x) &= \sum_{i=1}^{n} (-\tau \log q_i)\, x_i + \tau \sum_{i=1}^{n} x_i(\log x_i - 1) \\ &= \tau \sum_{i=1}^{n} \Big[x_i \log x_i - x_i \log q_i - x_i \Big] \\ &= \tau \sum_{i=1}^{n} \Big[x_i \log \tfrac{x_i}{q_i} - x_i \Big]. \end{aligned} \tag{A.2}$$

Define the negative-entropy function $\phi(x) := \sum_i x_i(\log x_i - 1)$ and its Bregman divergence $D_\phi(x\|q) := \sum_i \big[x_i \log(x_i/q_i) - x_i + q_i\big]$. Then (A.2) becomes

$$J(x) = \tau\, D_\phi(x\|q) - \tau \sum_i q_i. \tag{A.3}$$

Since $\tau \sum_i q_i$ is a constant with respect to x, minimizing $J(x)$ over the feasible set is equivalent to minimizing $D_\phi(x\|q)$.

KL projection when mass is fixed. When the constraints fix total mass $\sum_i x_i = N$ (as in population synthesis, transportation, and contingency tables), the Bregman divergence reduces to

$$D_\phi(x\|q) = \sum_i x_i \log \tfrac{x_i}{q_i} - N + \sum_i q_i = \mathrm{KL}(x\|q) + \sum_i q_i - N,$$

where $\mathrm{KL}(x\|q) = \sum_i x_i \log(x_i/q_i)$ is the (unnormalized) Kullback–Leibler divergence. The terms $\sum_i q_i - N$ are constant, so minimizing (2.3) is equivalent to the KL I-projection:

$$\min_{x \geq 0} \mathrm{KL}(x\|q) \quad \text{s.t. } Ax = b,\ \ell \leq x \leq u, \qquad q_i = e^{-c_i/\tau}. \tag{A.4}$$

This is exactly (2.4). When mass is not fixed (e.g., set covering), the full Bregman form (A.3) must be retained, but the IPF/Sinkhorn solution method remains structurally identical.

B Detailed proof of Theorem 2.1

While the theorem is intuitively natural—the entropy solution concentrates on the optimal vertex as $\tau \downarrow 0$, so MAP rounding must eventually recover it—a rigorous proof requires tracking this concentration through the full pipeline: first the Sinkhorn form and convergence of the dual potentials, then the exponential decay of mass on non-optimal edges, then the exact recovery argument for MAP rounding, and finally the induced objective convergence. Each step is straightforward in isolation, but all four checks are needed for the formal statement.

We give a detailed proof for the assignment problem, where the constraint polytope is the Birkhoff polytope $\mathcal{B}_n$ of doubly stochastic matrices. The extension to general problems with integral LP relaxations is stated as a corollary at the end.

Setup

Consider the balanced $n \times n$ assignment problem:

$$\min_{P \in \mathcal{P}_n} \langle C, P \rangle,$$

where $\mathcal{P}_n$ is the set of $n \times n$ permutation matrices and $C \in \mathbb{R}^{n \times n}_{\geq 0}$. Its LP relaxation over $\mathcal{B}_n$ has the same optimal value by Birkhoff's theorem. The entropy-regularized problem is

$$\min_{X \in \mathcal{B}_n} \langle C, X \rangle + \tau \sum_{i,j} X_{ij}(\log X_{ij} - 1).$$

We denote its unique solution (strict convexity) by X_τ.

Step 1: Sinkhorn form and dual potentials

The first-order conditions for X_τ yield the Sinkhorn scaling form

$$X_\tau(i,j) = \exp\Big(\frac{\alpha_{\tau,i} + \beta_{\tau,j} - C_{ij}}{\tau}\Big),$$

where $\alpha_\tau \in \mathbb{R}^n$ and $\beta_\tau \in \mathbb{R}^n$ are the dual potentials (Lagrange multipliers for the row and column constraints respectively), determined up to an additive constant by the doubly stochastic requirement. These are computed iteratively by Sinkhorn scaling.

By standard Γ-convergence results for entropic optimal transport [6, 27], $(\alpha_\tau, \beta_\tau) \to (\alpha^*, \beta^*)$ uniformly as $\tau \to 0$, where (α^*, β^*) are optimal dual variables for the LP relaxation satisfying

$$\alpha_i^* + \beta_j^* \leq C_{ij} \quad \forall i,j, \quad \text{with equality if } (i,j) \in P^*.$$

Step 2: Exponential concentration on optimal edges

For edges $(i,j) \in P^*$, complementary slackness gives $\alpha_i^* + \beta_j^* = C_{ij}$, so

$$X_\tau(i,j) = \exp\Big(\frac{\alpha_{\tau,i} + \beta_{\tau,j} - C_{ij}}{\tau}\Big) \to \exp(0) = 1 \quad \text{as } \tau \to 0.$$

More precisely, since $(\alpha_\tau, \beta_\tau) \to (\alpha^*, \beta^*)$ uniformly, the exponent is $O(\tau \log n)$, giving $X_\tau(i,j) = 1 - O(\tau \log n)$ for $(i,j) \in P^*$.

For edges $(i,j) \notin P^*$, let $\delta := \min\{C_{ij} - (\alpha_i^* + \beta_j^*) : (i,j) \notin P^*\} > 0$ (positive by strict complementarity, which follows from uniqueness of P^*). Then

$$X_\tau(i,j) = \exp\Big(\frac{\alpha_{\tau,i} + \beta_{\tau,j} - C_{ij}}{\tau}\Big) \leq \exp(-\delta/(2\tau))$$

for all τ small enough that $|\alpha_{\tau,i} - \alpha_i^*| + |\beta_{\tau,j} - \beta_j^*| \leq \delta/2$.

Since row i of X_τ sums to 1, mass must concentrate on optimal edges. For $(i,j) \in P^*$:

$$X_\tau(i,j) \geq 1 - \sum_{k \neq j} X_\tau(i,k) \geq 1 - n\exp(-\delta/(2\tau)) \to 1.$$

Step 3: MAP rounding recovers P^*

MAP rounding selects $\hat{P} \in \arg\min_{P \in \mathcal{P}_n} \sum_{i,j} P_{ij}(-\log X_\tau(i,j))$. We show $\hat{P} = P^*$ for τ sufficiently small.

For $(i,j) \in P^*$:

$$-\log X_\tau(i,j) \leq -\log\bigl(1 - ne^{-\delta/(2\tau)}\bigr) \leq 2ne^{-\delta/(2\tau)},$$

using $-\log(1-u) \leq 2u$ for $u \leq 1/2$, valid for $\tau < \delta/(2\log(2n))$.

For $(i,j) \notin P^*$:

$$-\log X_\tau(i,j) \geq \delta/(2\tau).$$

Any permutation $Q \neq P^*$ must use at least one non-optimal edge, incurring MAP cost at least

$$\frac{\delta}{2\tau} - (n-1) \cdot 2ne^{-\delta/(2\tau)}.$$

The cost of P^* under MAP is at most $n \cdot 2ne^{-\delta/(2\tau)}$. The excess cost of any $Q \neq P^*$ over P^* is therefore at least

$$\frac{\delta}{2\tau} - (2n^2 - 1) \cdot 2ne^{-\delta/(2\tau)}.$$

For $\tau < \delta/(4\log n)$, the first term dominates and this excess is strictly positive, so $\hat{P} = P^*$. □

Step 4: Objective convergence

The MAP objective satisfies

$$\langle C, P_\tau^* \rangle \leq \langle C, P^* \rangle + \tau\, n \log n,$$

since the entropy penalty of P^* is at most $n \log n$ (each entry is at most n^{-1} by doubly stochasticity). As $\tau \to 0$, $\langle C, \hat{P} \rangle \to \langle C, P^* \rangle$. □

Extension to general integral polyhedra

Corollary B.1 *Let $\mathcal{P} = \{x \geq 0 : Ax = b\}$ be any polytope with only integer vertices, and let x^* be the unique LP optimum. Under the same strict complementarity assumption, the MAP-rounded solution x_τ^* satisfies $\langle c, x_\tau^* \rangle \to \langle c, x^* \rangle$ as $\tau \to 0$, and $x_\tau^* = x^*$ for all $\tau < \delta/K$ with problem-dependent K.*

Proof The Sinkhorn form $x_{\tau,i} = \exp((a_i^\top \lambda_\tau - c_i)/\tau)$ (for active coordinates) is the general-polytope analogue of Step 1 above. The concentration and MAP recovery arguments of Steps 2 and 3 carry over verbatim with reduced cost $r_i = c_i - a_i^\top \lambda^*$ playing the role of $C_{ij} - (\alpha_i^* + \beta_j^*)$, and $\delta = \min\{r_i : x_i^* = 0\} > 0$ by strict complementarity. □

C Detailed proof of Theorem 2.2

We prove each part of Theorem 2.2 in turn. The argument parallels standard analyses of entropic regularization in optimal transport [6, 27]. We use the notation of Theorem 2.2: $\mathcal{P} = \{x \geq 0 : Ax = b\}$ is a lattice polytope, $x^\star$ is the unique LP optimum, $\lambda^\star$ is an optimal dual with slack $\delta = \min\{c_i - a_i^\top \lambda^\star : x_i^\star = 0\} > 0$, and $r_i := c_i - a_i^\top \lambda^\star$ denotes the reduced cost.

Step 1: Stationarity and dual convergence. The first-order conditions for x_τ are:

$$c_i + \tau \log x_{\tau,i} = a_i^\top \lambda_\tau + s_{\tau,i}, \quad s_{\tau,i}\, x_{\tau,i} = 0, \quad s_{\tau,i} \geq 0.$$

For active coordinates ($x_{\tau,i} > 0$, so $s_{\tau,i} = 0$):

$$x_{\tau,i} = \exp\Big(\tfrac{a_i^\top \lambda_\tau - c_i}{\tau}\Big).$$

By Γ-convergence of the entropy-regularized problem to the LP as $\tau \to 0$ [6], we have $\lambda_\tau \to \lambda^\star$ uniformly; let $\varepsilon(\tau) := \|\lambda_\tau - \lambda^\star\|_\infty \to 0$.

Step 2: Part (c) — exponential concentration. For nonbasic indices ($x_i^\star = 0$, so $r_i \geq \delta$):

$$a_i^\top \lambda_\tau - c_i = -r_i + a_i^\top (\lambda_\tau - \lambda^\star) \leq -\delta + \|a_i\|_1\, \varepsilon(\tau).$$

For τ small enough that $\|a_i\|_1\, \varepsilon(\tau) \leq \delta/2$:

$$x_{\tau,i} \leq \exp(-\delta/(2\tau)) =: C_1 e^{-\delta/(2\tau)},$$

establishing the pointwise bound in part (c) with $C_1 = 1$. Since $Ax_\tau = b = Ax^\star$, we have $A(x_\tau - x^\star) = 0$. Using $x_{\tau,i} \geq 0$ and the bound on nonbasic components:

$$\|x_\tau - x^\star\|_1 \leq 2 \sum_{x_i^\star = 0} x_{\tau,i} \leq 2n\, C_1 e^{-\delta/(2\tau)} =: C_2 e^{-\delta/(2\tau)},$$

completing part (c).

Step 3: Part (a) — objective gap.

$$\begin{aligned} c^\top x_\tau - c^\top x^\star = c^\top (x_\tau - x^\star) &= (c - A^\top \lambda^\star)^\top (x_\tau - x^\star) + (\lambda^\star)^\top A(x_\tau - x^\star) \\ &= r^\top (x_\tau - x^\star), \end{aligned}$$

using $A(x_\tau - x^\star) = 0$. Since $r_i = 0$ for basic indices and $r_i \geq 0$ for nonbasic indices:

$$c^\top x_\tau - c^\top x^\star = \sum_{x_i^\star = 0} r_i (x_{\tau,i} - x_i^\star) \leq \|r\|_\infty\, \|x_\tau - x^\star\|_1 \leq \|r\|_\infty C_2\, e^{-\delta/(2\tau)} = O(\tau),$$

where the last equality uses $e^{-\delta/(2\tau)} = O(\tau)$ as $\tau \to 0$ (since $e^{-\delta/(2\tau)}$ decays faster than any polynomial). This gives part (a) with constant $C_2 \|r\|_\infty$.

Step 4: Part (b) — exact recovery by MAP. MAP rounding selects $\hat{x} \in \arg\min_{v \in \mathrm{vert}(\mathcal{P})} \sum_i v_i(-\log x_{\tau,i})$.

For basic indices ($x_i^\star > 0$): $x_{\tau,i} = O(1)$, so $-\log x_{\tau,i} = O(1)$.

For nonbasic indices: $x_{\tau,i} \leq C_1 e^{-\delta/(2\tau)}$, so $-\log x_{\tau,i} \geq \delta/(2\tau) - \log C_1$.

Any vertex $v \neq x^\star$ must have $v_i > 0$ for at least one nonbasic index i (since $x^\star$ is the unique LP optimum, no other vertex shares its support). The excess MAP cost of v over $x^\star$ is at least

$$\frac{\delta}{2\tau} - \log C_1 - K_0,$$

where K_0 is an $O(1)$ bound on the contribution of basic indices. For $\tau < \delta/(2(K_0 + \log C_1 + 1))$, this excess is strictly positive, forcing $\hat{x} = x^\star$. Setting $K = 2(K_0 + \log C_1 + 1)$ gives part (b). □